

MAST32001 Computational Statistics

Exercises 2

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Instructions

For each exercise:

- reason about the method before running the code;
- implement the requested simulation yourself where appropriate;
- check whether the simulated output behaves as theoretically expected;
- distinguish statistical variation from Monte Carlo variation;
- clearly document your reasoning and R code.

You may use base R and the functions introduced in the lecture. If a task asks you to generate a distribution by a particular method, do not replace that step by the corresponding built-in `r...()` function.

Submission

Solutions must be submitted in **Moodle no later than 15.09.2026 at 12:00**.

Please submit your solutions in the **submission area corresponding to your exercise group**. Make sure that you submit to the correct exercise group.

Exercise points

Exercise points are awarded only in connection with the corresponding exercise session.

To receive the point(s) for a task, you must:

1. submit your solution by the deadline;
2. attend the exercise session in which the task is discussed; and
3. be willing to **present, explain, and discuss your solution** during the exercise session if asked.

Submitting a solution without attending the corresponding exercise session does **not** earn exercise points.

Your solution does not need to be perfect in order to be discussed. The purpose of the exercise sessions is also to compare approaches, identify difficulties, and learn from different solutions.

Use of generative AI

The use of **generative AI tools to solve the exercise tasks is not permitted**.

The purpose of the exercises is to develop your own ability to translate statistical ideas into algorithms, implement them in R, investigate numerical behaviour, and explain your reasoning.

The submitted solutions, including the reasoning and R code, must therefore reflect **your own work**.

Exercise 1 — Generate it, then check it

Let X have the Pareto distribution with scale $x_0 = 2$ and shape $\alpha = 3$, with cdf

$$F(x) = 1 - \left(\frac{x_0}{x}\right)^\alpha, \quad x \geq x_0.$$

Tasks

1. Derive the inverse cdf $F^{-1}(u)$.
2. Write a function

```
rpareto_inv <- function(n, x0, alpha) {  
  # your code  
}
```

that generates Pareto random variables using only `runif()` and the inverse transformation method.

3. Set the seed to 20260812 and generate $n = 5000$ observations with $x_0 = 2$ and $\alpha = 3$.
4. Check the generated sample graphically in at least the following ways:
 - make a histogram and overlay the theoretical Pareto density;
 - make a Q-Q plot comparing the empirical quantiles with theoretical Pareto quantiles;
 - compare the empirical cdf with the theoretical cdf.

You may define the density and quantile functions yourself.

5. The Pareto distribution has

$$E(X) = \frac{\alpha x_0}{\alpha - 1}, \quad \alpha > 1.$$

Compare the sample mean with the theoretical mean.

6. Repeat the simulation with $\alpha = 1.2$. What changes in the plots and in the stability of the sample mean? Explain why checking a random-number generator from one finite sample can never prove that the generator is correct.
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Exercise 2 — Monte Carlo integration and Monte Carlo error

Consider

$$I = \int_0^1 \frac{1}{1+x^2} dx.$$

The exact value is $I = \pi/4$.

Tasks

1. Explain why, for $U \sim U(0, 1)$,

$$I = E \left[\frac{1}{1+U^2} \right].$$

2. For $m = 100, 1000, 10000, 100000$, estimate I by Monte Carlo. For every m , report:
 - the Monte Carlo estimate;
 - its absolute error relative to $\pi/4$;
 - an estimated Monte Carlo standard error (MCSE).
3. Compare the Monte Carlo result for $m = 100000$ with R's deterministic numerical integration via `integrate`. Why should `integrate()` generally be preferred for an easy one-dimensional integral such as this one?
4. The Monte Carlo standard error decreases at rate $m^{-1/2}$. Approximately how many simulations would be required to make the MCSE ten times smaller?
5. Repeat the complete Monte Carlo experiment 500 times with $m = 1000$. Standardize each estimate using its estimated MCSE,

$$Z_b = \frac{\hat{I}_b - I}{\widehat{\text{MCSE}}_b}.$$

Make a histogram and a normal Q-Q plot of the Z_b . What feature of Monte Carlo theory are you checking?

Exercise 3 — Simulating size and power of a test

Suppose $X_1, \dots, X_n$ are independent observations from

$$N(\mu, 1),$$

and consider testing

$$H_0 : \mu = 0 \quad \text{against} \quad H_1 : \mu > 0.$$

Use the test statistic

$$Z = \sqrt{n} \bar{X}$$

and reject H_0 when

$$Z > z_{0.95}.$$

Take $n = 25$.

Tasks

1. Simulate $B = 10000$ data sets under H_0 , apply the test to every data set, and estimate the rejection probability.
2. What should this probability be theoretically? Report a Monte Carlo standard error for your estimated rejection probability. Is your result compatible with the nominal significance level?
3. Repeat the simulation under the alternative $\mu = 0.4$. The rejection probability is now an estimate of the **power**.
4. Derive the theoretical power for a general value of μ , and compare your simulated power at $\mu = 0.4$ with the theoretical value.
5. Estimate the power for

$$\mu = 0, 0.1, 0.2, \dots, 0.8$$

using $B = 5000$ simulations at each value. Plot the estimated power curve and add the theoretical power curve.

6. Explain the roles of:
 - the significance level;
 - the effect size μ ;
 - the sample size n ;
 - the number of Monte Carlo repetitions B .

In particular, explain why increasing B makes the estimated power more precise but does **not** make the statistical test itself more powerful.

Exercise 4 — Simulating and diagnosing a five-dimensional normal distribution

Let

$$X \sim N_5(\mu, \Sigma),$$

where

$$\mu = \begin{pmatrix} 2 \\ -1 \\ 0 \\ 3 \\ 1 \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} 4 & 1.2 & 0.8 & 0.0 & 1.0 \\ 1.2 & 2.0 & 0.3 & 0.6 & 0.2 \\ 0.8 & 0.3 & 1.5 & 0.4 & 0.5 \\ 0.0 & 0.6 & 0.4 & 3.0 & 1.1 \\ 1.0 & 0.2 & 0.5 & 1.1 & 2.5 \end{pmatrix}.$$

Tasks

1. Check that Σ is positive definite.
2. Generate $n = 2000$ observations from $N_5(\mu, \Sigma)$ **without using a multivariate-normal simulation package**.

Hint: If

$$\Sigma = R^\top R$$

is obtained from `R <- chol(Sigma)` and Z is an $n \times 5$ matrix of independent $N(0, 1)$ observations, then think about the covariance matrix of

```
Z %*% R
```

and then add the mean vector.

3. Compare the sample mean vector and sample covariance matrix with μ and Σ .
4. Make a scatterplot matrix using `pairs()`. Comment on:
 - the centre of the clouds;
 - the direction of association;
 - the strength of association;
 - the approximately elliptical shape of the bivariate projections.

Which entries of Σ help explain what you see?

5. **Whiten the centred observations.**

Hint: with `R <- chol(Sigma)`, your simulated centred data have the form

```
Z %*% R
```

so multiplying from the right by `solve(R)` reverses the transformation:

```
X_white <- X_centered %*% solve(R)
```

Check the sample covariance matrix of the whitened data and make another scatterplot matrix. What has changed?

6. For each observation compute

$$D_i^2 = (X_i - \mu)^\top \Sigma^{-1} (X_i - \mu),$$

the **squared Mahalanobis distance** from the true mean.

Hint: after correct whitening,

$$D_i^2 = \|X_{i,\text{white}}\|_2^2.$$

Thus you can compute all squared distances using

```
rowSums(X_white^2)
```

7. If $X \sim N_5(\mu, \Sigma)$, then theoretically

$$D^2 \sim \chi_5^2.$$

Make a Q-Q plot of the observed squared Mahalanobis distances against theoretical χ_5^2 quantiles. Add the line $y = x$.

8. Summarize the evidence from the mean, covariance matrix, scatterplots, whitening, and Mahalanobis-distance Q-Q plot. Does the simulated sample look consistent with the requested multivariate normal distribution? Which of these checks examine only first and second moments, and which provide additional distributional information?

Exercise 5 — Importance sampling for a rare tail probability

Let $Z \sim N(0, 1)$. We want to estimate the small probability

$$p = P(Z > 5).$$

A direct Monte Carlo estimator is

$$\hat{p}_{\text{direct}} = \frac{1}{m} \sum_{i=1}^m I(Z_i > 5), \quad Z_i \sim N(0, 1).$$

For a rare event, most simulated observations contribute exactly zero.

An alternative is **importance sampling**. Generate instead

$$Y_i \sim N(5, 1)$$

and use

$$p = E_g \left[I(Y > 5) \frac{f(Y)}{g(Y)} \right],$$

where f is the $N(0, 1)$ density and g is the $N(5, 1)$ density.

Tasks

1. Use $m = 100000$ and estimate $P(Z > 5)$ by direct Monte Carlo.
2. Count how many simulated observations exceed 5. Explain why the direct estimator is difficult to use for this probability.
3. With the same m , generate $Y_i \sim N(5, 1)$ and compute the importance weights

$$w_i = \frac{f(Y_i)}{g(Y_i)}.$$

Estimate p using

$$\hat{p}_{\text{IS}} = \frac{1}{m} \sum_{i=1}^m I(Y_i > 5) w_i.$$

4. Compare both estimates with

```
pnorm(5, lower.tail = FALSE)
```

5. Estimate the MCSE of both estimators from the standard deviation of their individual Monte Carlo contributions divided by $\sqrt{m}$.
6. Repeat both methods $B = 200$ times using $m = 10000$ per repetition. Compare the distributions of the resulting estimates using boxplots or histograms.
7. Explain why sampling from $N(5, 1)$ can make the rare-event calculation much more efficient. Why would choosing an importance distribution centred very far from the relevant region also be potentially problematic?